

JavaScript Syntax

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JavaScript Values

The JavaScript syntax defines two types of values:

- **Fixed values** are called **Literals**.
- **Variable values** are called **Variables**.

JavaScript Literals

The most important syntax rules for **literals** (fixed values) are:

Numbers are written with or without decimals:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Numbers</h4>
<p>Number can be written with or without decimals:</p>
<p id="demo1"></p>
<p id="demo2"></p>

<script>
document.getElementById("demo1").innerHTML = 10.50;
document.getElementById("demo2").innerHTML = 1001;
</script>

</body>
</html>
```

Strings are text, written within double or single quotes:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Strings</h4>
<p>Strings can be written with double or single quotes:</p>
<p id="demo3"></p>
<p id="demo4"></p>

<script>
document.getElementById("demo3").innerHTML = "John Doe";
document.getElementById("demo4").innerHTML = 'John Doe';
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

JavaScript Variables

Variables are containers for **storing data values**.

Variables must be **identified** with **unique names**.

The `let` and `const` keywords create variables:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Variables</h4>
<p id="demo"></p>

<script>
let x;
x = 6;
document.getElementById("demo").innerHTML = x;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

JavaScript Identifiers

An **identifier** is the **name** given to a variable.

Rules for identifiers:

- Names can contain letters, digits, underscores, and dollar signs.
- Names must begin with a letter.
- Names can also begin with `$` and `_`.
- Names are case sensitive (y and Y are different variables).

- Reserved words (like JavaScript keywords) cannot be used as names.

JavaScript Operators

JavaScript **assignment operators** (`=`) assign values to variables:

JavaScript uses **arithmetic operators** (`+` `-` `*` `/`) to **compute** values:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Operators</h4>
<p id="demo1"></p>
<p id="demo2"></p>

<script>
let a, b, c;
a = 5;
b = 2;
c = a + b;
document.getElementById("demo1").innerHTML = "Value of a + b = " + c;
document.getElementById("demo2").innerHTML = "Value of a * b = " + (a * b);
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

JavaScript Expressions

An expression is a combination of values, variables, and operators, which computes to a value:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Expressions</h4>
<p id="demo1"></p>
<p id="demo2"></p>
<p id="demo3"></p>

<script>
document.getElementById("demo1").innerHTML = (5 + 6) * 10;
let x = 5;
document.getElementById("demo2").innerHTML = x * 10;
document.getElementById("demo3").innerHTML = "John" + " " + "Doe";
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

JavaScript is Case Sensitive

JavaScript identifiers are **case sensitive**.

The variables `lastName` and `lastname` , are different variables:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Syntax</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript is Case Sensitive</h4>
<p id="demo1"></p>
<p id="demo2"></p>

<script>
let lastName = "Doe";
let lastname = "Smith";
document.getElementById("demo1").innerHTML = lastName;
document.getElementById("demo2").innerHTML = lastname;
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

JavaScript and Camel Case

Historically, programmers have used different ways of joining multiple words into one variable name:

- **Hyphens:** first-name, last-name. Hyphens are **not allowed** in JavaScript. They are reserved for subtractions.
- **Underscore:** first_name, last_name.
- **Upper Camel Case (Pascal Case):** FirstName, LastName.
- **Lower Camel Case:** firstName, lastName.

JavaScript programmers tend to use **lower camel case**.

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Syntax](#)